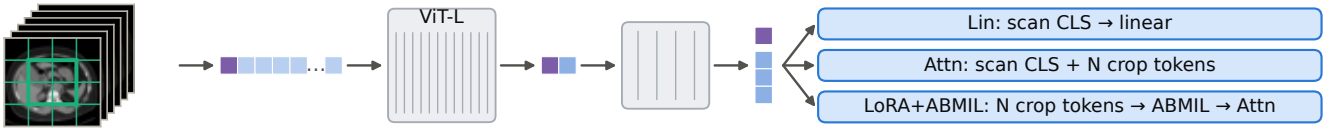
(a) SPECTRE: crops, per-crop ViT-L, feature combiner



512²×256, 0.5×0.5×1 mm, 4³ = 64 crops (train: 2³ = 8)

per 128²×64 crop: 16×16×8 patches, 512 + CLS tokens

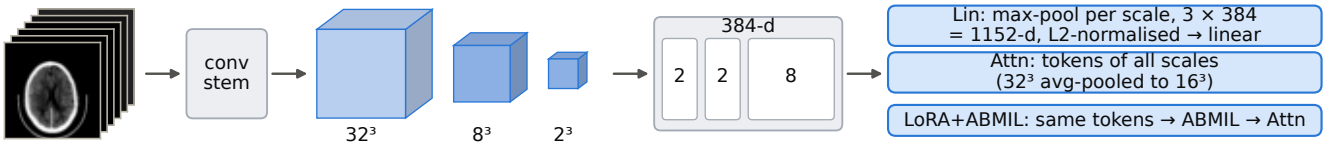
24 blocks, 1080-d, per crop

[CLS ‖ mean], 2160-d

combiner: 4 blocks over crops

scan CLS + N crop tokens

(b) Pillar-0 (head CT): multi-window input, conv stem, three-stage multi-scale encoder



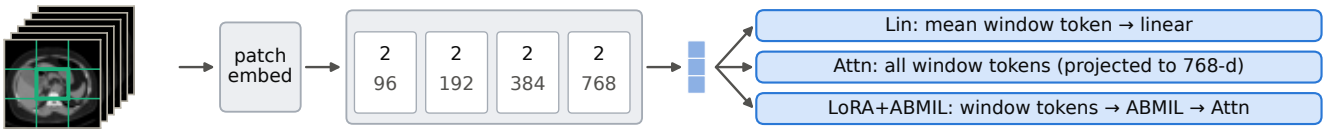
128×256×256, 0.5×0.5×1.25 mm; 11 channels: 10 HU windows + min–max

8×8×4 patches, 32³ grid

tokens at 3 scales (max-pool by 4 per axis), attention within and across scales

3 stages of 2, 2, 8 blocks; scales 32³, 8³, 2³ leave after stages 1, 2, 3; no CLS token

(c) VoCo: 96³ windows, four-stage Swin encoder, one token per window



RAS, 1.5 mm, HU −175…250, 96³ windows (train: ≤ 16)

2³ patches, 96 × 48³

2 blocks per stage, widths 96, 192, 384, 768 (shifted 7³ windows); stage outputs after merging: 192, 384, 768, 1536 at strides 4–32

window token: means of stage-2–4 outputs, concatenated: 384 + 768 + 1536 = 2688-d

(d) What each classification method trains

		early blocks			late blocks					encoder parameters updated			
		patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE	Pillar-0	VoCo
Lin									pooled → linear	8000	0	0	0	0
Attn									attentive probe	8000	0	0	0	0
LoRA									attentive probe	8000	0.59 M	2.4 M	0.55 M*	0.18 M
LoRA$_{late}$									attentive probe	8000	0.20 M	1.0 M	0.25 M*	0.10 M
LoRA+ABMIL									ABMIL → Attn	8000	0.59 M	2.4 M	0.55 M*	0.18 M
DoRA	(+ m)		m			m			attentive probe	8000	0.62 M	2.5 M	0.57 M*	0.19 M
rsLoRA	(α/√r)								attentive probe	8000	0.59 M	2.4 M	0.55 M*	0.18 M
Lin→LoRA						2			linear	4000 + 4000	0.20 M	1.0 M	0.25 M*	0.10 M†
Attn→LoRA						2			attentive probe	4000 + 4000	0.20 M	1.0 M	0.25 M*	0.10 M†
Attn-only FT									attentive probe	8000	28 M	131 M	11 M*	6.5 M
Full FT									attentive probe	8000	87 M	397 M	45 M*	74 M

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full

W_{qkv}: fused qkv (TAP-CT, VoCo) or q and kv (SPECTRE, Pillar-0); Attn-only FT trains the whole attention module (with biases;

VoCo: also the relative position bias). Late blocks: TAP-CT 9–12, SPECTRE 17–24 and its combiner, the last stage of Pillar-0 (8 of 12 blocks) and of VoCo (2 of 8). CVPR26 joint models also ran 26 000 steps (13 000 + 13 000). Head parameters for one label (TAP-CT / SPECTRE / Pillar-0 / VoCo): linear 769 / 1081 / 1153 / 2689; attentive probe 14.2 / 28.0 / 31.9 / 16.2 M. †: same late blocks as LoRA$_{late}$.

*Pillar-0: the per-scale blocks of scales that have already left the encoder, and the text projection, are also wrapped or unfrozen but never run and receive no gradient; counting them: LoRA 1.1 M, LoRA$_{late}$ 0.74 M, Attn-only FT 21 M, Full FT 94 M.